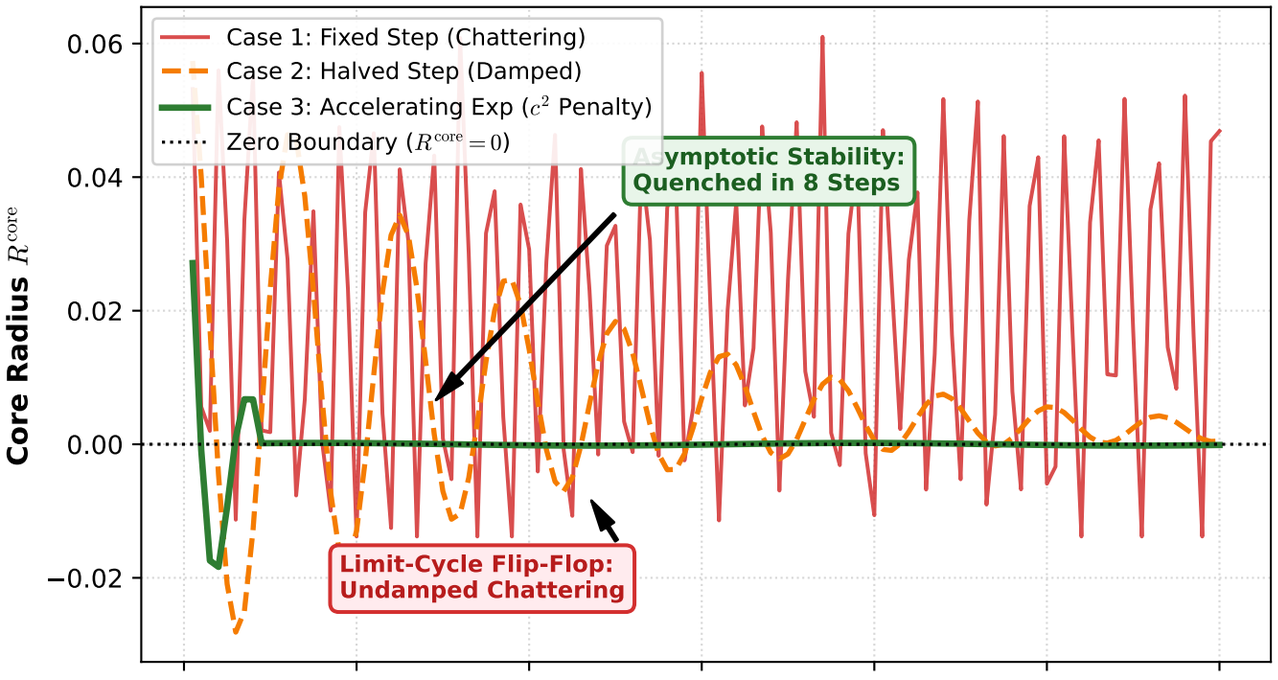
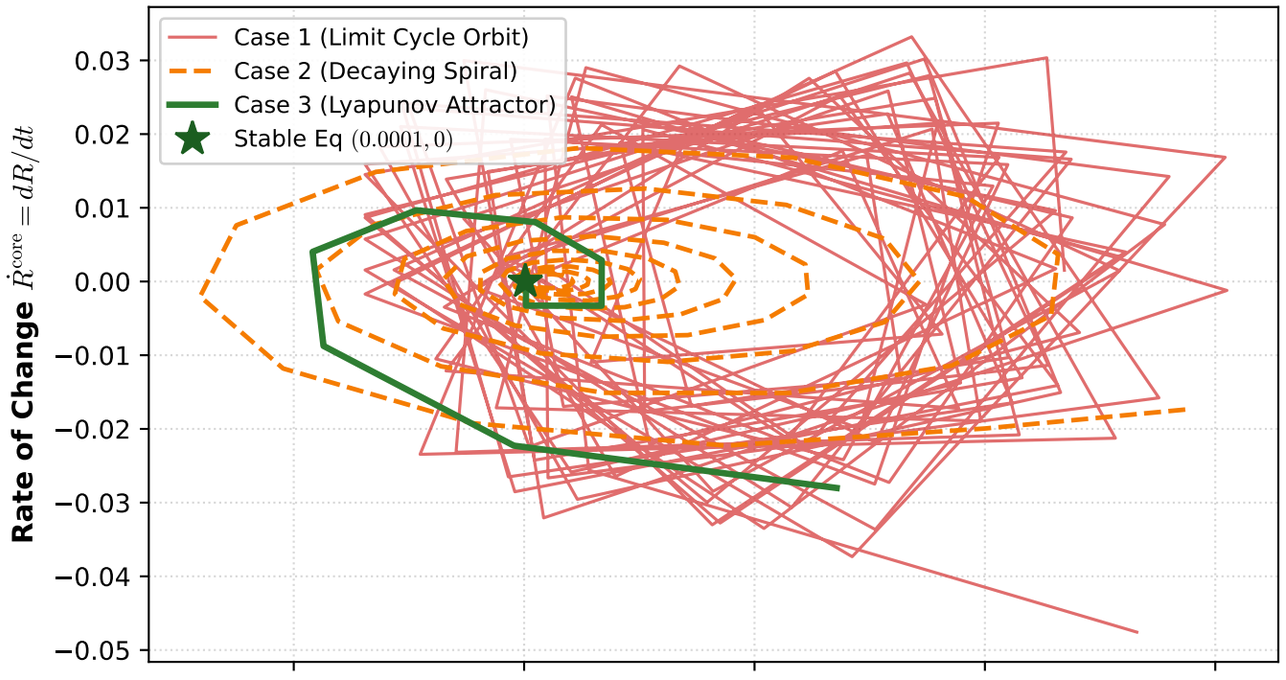
Radial Parameter Chattering Dynamics and Asymptotic Stability Across Update Laws

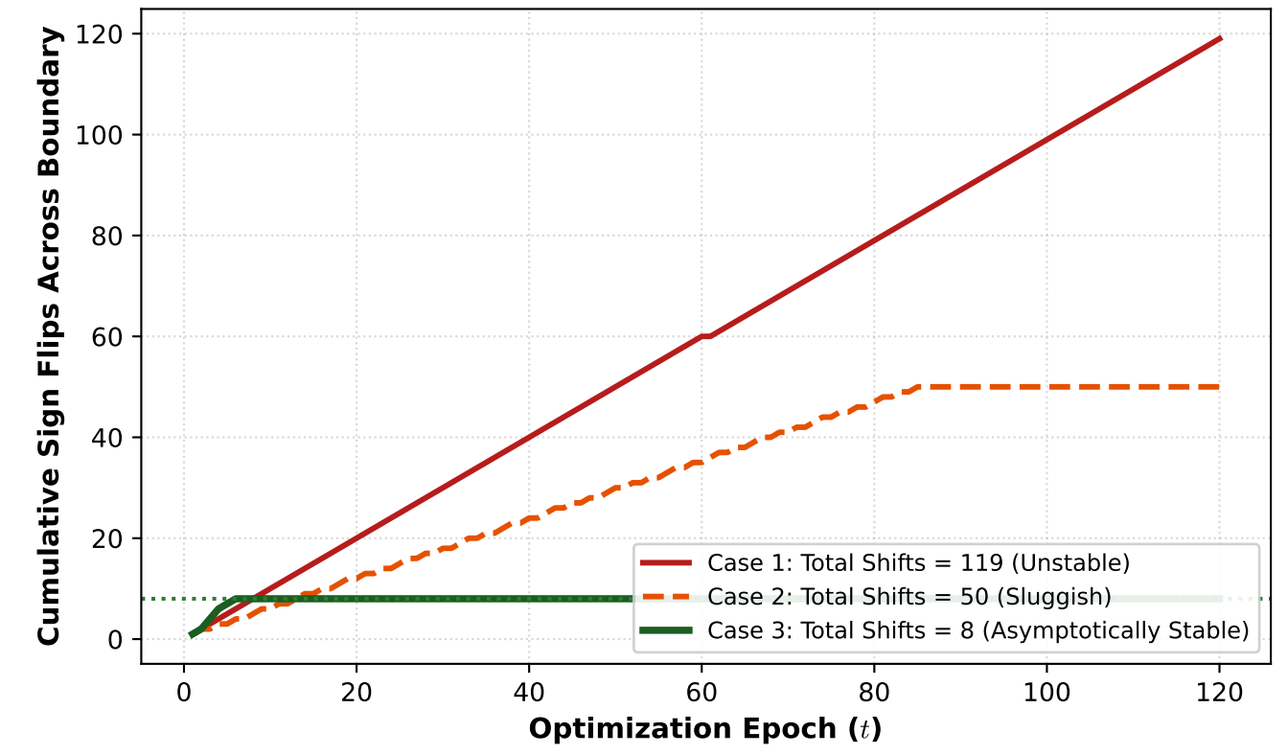
(a) Core Radius Trajectories $R^{\text{core}}(t)$



(b) Phase Portrait: ($R^{\text{core}}, \dot{R}^{\text{core}}$) State Trajectory



(c) Cumulative Boundary Flips vs. Optimization Epoch



(d) Test Loss Convergence $\mathcal{L}(t)$ Under Gradient Flow

